CLASS 7 — MATHS · FORMULA SHEET

Chapter 6: **Number Play** | Quick revision, one page

Odd and even

- Even number: $2n$. Odd number: $2n + 1$.
- Even $\pm$ even $=$ even; odd $\pm$ odd $=$ even; even $\pm$ odd $=$ odd.
- Even $\times$ any whole number $=$ even; odd $\times$ odd $=$ odd.

Digits and reversals

- Two-digit number with digits a, b: $10a + b$; its reverse: $10b + a$.
- Sum $= 11(a + b)$, so the sum is divisible by 11; difference $= 9(a - b)$, divisible by 9.
- Palindrome: reads the same both ways.

Divisibility

- By 3 (or 9): digit sum divisible by 3 (or 9). By 2, 5, 10: look at the last digit.
- By 4: last two digits; by 8: last three digits; by 11: alternating digit sum.

Consecutive numbers

- Three consecutive numbers: $(n - 1) + n + (n + 1) = 3n$. Two consecutive numbers: $n + (n + 1) = 2n + 1$ (odd).
- Sum of the first n odd numbers $= n^2$.

Number sequences

- Collatz: even $\to n \div 2$; odd $\to 3n + 1$; ends in the loop $4, 2, 1$.
- Kaprekar routine: (largest arrangement) $-$ (smallest arrangement); 4-digit numbers (not all digits equal) reach 6174.

MASTERMIND